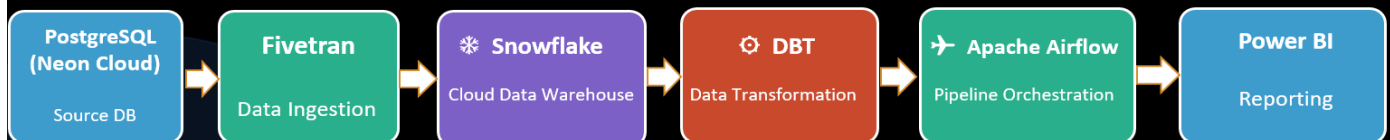


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Professional IT Training | Data Engineering | Cloud Computing

Snowflake Combo Online Training Build a High-Paying Cloud Data Engineering Careers in 2026



45 Sessions

6 Cloud Tools

3 Live Projects

Career Ready

TRAINER: HARIA | Founder & Chief Trainer, 15+ Yrs | Rishab Informatica Group | +91 - 8970853557

SNOWFLAKE DATA ENGINEERING COMBO ONLINE TRAINING

COURSE CONTENT

❄️ Snowflake + ⚙️ DBT + ✈️ Apache Airflow
(Data Ingestions from ADF , AWS Etc,.)

The Most Demanded Modern Data Engineering Stack of 2026

TRAINER

HARI.A

Founder & Chief Trainer

Rishab Informatica Group

15+ Years Industry Experience

EXPERTISE

- Snowflake
- DBT (Data Build Tool)
- Apache Airflow
- Informatica IICS
- SQL | Python | Cloud (AWS / Azure / GCP)

47

Total Sessions

1.5–2 hrs

Per Session

3

Technologies

3

Live Projects

OUR TRAINING PHILOSOPHY

At Rishab Informatica Group, we don't just teach tools.

We train students to crack interviews, survive real projects, and grow confidently in IT careers.

HOW OUR TRAINING WORKS

- Interview Preparation from Day 1
- Real Interview Exposure from Session 25
- Multiple Mock Interviews with Feedback
- Concept + Scenario + Project-Based Learning
- Complete Support Till Job Placement
- FREE Job Support After Placement

WHAT MAKES US DIFFERENT

- Max 25 Students Per Batch — Personal Attention
- No Mute Policy — 100% Interactive Sessions
- Recorded Videos via LMS, Android & iOS App
- 2 Real-Time Projects Included
- Interview-Oriented from Day 1
- Lifetime Mentorship & Doubt Clearance

PROVEN SUCCESS FLOW

Day 1	Day 20	Day 23	Day 25+	Day 35	Day 40-47
Interview Orientation How interviewers think & how to answer confidently	Resume Ready Naukri & LinkedIn profile creation	Mock Interview 1 Based only on completed sessions	Real Interviews Start attending actual interviews	Mock Interview 2 Advanced + Scenario-based questions	Job Cracked! Expected placement + FREE job support

COURSE OVERVIEW

❄️ PART A	⚙️ PART B	➔ PART C
SNOWFLAKE	DBT	APACHE AIRFLOW
Days 1 – 26 26 Sessions Data Warehousing & Cloud Analytics	Days 27 – 36 10 Sessions Modern ELT Transformations	Days 37 – 47 11 Sessions Pipeline Orchestration & Scheduling

TECHNOLOGY STACK

TOOL	PURPOSE IN THIS COURSE	REAL WORLD ROLE
❄️ Snowflake	Store, query and manage data in cloud	Data Warehouse — holds all processed data
⚙️ DBT	Transform raw data into clean analytics models	Transformation Layer — SQL-based data modeling
➔ Airflow	Schedule & orchestrate entire data pipelines	Orchestration Layer — automates when & what runs

THE COMPLETE DATA PIPELINE YOU WILL BUILD

Sources → Fivetran → Apache Airflow → Snowflake → DBT → Power Bi
(Orchestrate) (Store) (Transform)



PART A: SNOWFLAKE

Data Warehousing & Cloud Analytics

Days 1 – 26 | 26 Sessions

DAY
1

Introduction & Interview Orientation

- What is Data Engineering & the Modern Data Stack
- Why Snowflake? Market Demand & Career Scope in 2026
- Snowflake Architecture — Multi-cluster Shared Data
- Snowflake vs Traditional Databases & Data Warehouses
- Full Interview Q&A Introduction — How Interviewers Think

DAY
2

Account Setup & Tools

- Snowflake Trial Account Creation Step by Step
- Snowsight Web UI — Navigation & Features
- SnowSQL CLI Installation & Configuration
- Partner Connect Overview
- Hands-on: First Queries & Exploring the Interface

DAY
3

Roles, Users & Security

- Creating Roles & Users in Snowflake
- Role Hierarchy & Privilege Management
- GRANT & REVOKE Commands
- Resource Monitor Basics — Budget Controls
- Interview Q&A: Security & Access Control

DAY
4

Virtual Warehouses

- Warehouse Creation & Size Types (X-Small to 6X-Large)
- Auto Suspend & Auto Resume Configuration
- Credit Usage & Cost Monitoring
- Multi-Cluster Warehouses for Concurrency
- Performance Tuning with Warehouse Sizing

DAY
5

Databases & Schemas

- Database & Schema Creation Best Practices
- Snowflake Storage Architecture Deep Dive
- Micro-partitions — How Snowflake Stores Data
- Metadata & Automatic Clustering Overview
- Interview Q&A: Architecture Questions

DAY
6

Tables & Constraints

- Table Types: Permanent, Transient & Temporary
- DDL Operations — CREATE, ALTER, DROP, TRUNCATE
- DML Operations — INSERT, UPDATE, DELETE, MERGE
- Constraints — Primary Key, Foreign Key, Unique, NOT NULL
- Iceberg Tables — Advanced Awareness for 2026

DAY
7

Joins & Set Operators

- All Join Types — INNER, LEFT, RIGHT, FULL OUTER
- CROSS JOIN & SELF JOIN with Real Examples
- Set Operators — UNION, UNION ALL, INTERSECT, EXCEPT
- Views Overview — When & Why to Use
- Practical SQL Lab with Real Data

DAY
8

Data Loading — Local Files

- Load CSV, JSON, XML from Local Machine
- PUT Command — Stage Files from Local
- COPY INTO Command — Load from Stage to Table
- File Format Objects — CSV, JSON, Parquet, ORC
- Error Handling — ON_ERROR Options & Best Practices

DAY
9

Data Loading — Cloud Storage

- AWS S3 Integration — Setup & Configuration
- Load CSV, JSON, Parquet from S3 Bucket
- Azure Blob Storage Integration Overview

- Google Cloud Storage Integration Overview
- Snowpipe Introduction — Auto Ingestion Concept

DAY
10

Stages & CLI Operations

- Internal Stages — User Stage, Table Stage, Named Stage
- External Stages — S3, Azure, GCS
- Named Stages with File Format Objects
- SnowSQL CLI — Day to Day Operations
- Bulk Loading Best Practices & Performance Tips

DAY
11

Time Travel & Fail Safe

- Time Travel — BEFORE & AT Keywords
- Querying Historical Data with Time Travel
- Restoring Dropped Tables, Schemas & Databases
- Query History in Snowsight UI
- Fail Safe — 7 Day Recovery Concept & Limitations

DAY
12

Zero Copy Cloning

- Zero Copy Clone — How it Works Internally
- Clone Tables, Schemas & Databases Instantly
- Use Cases — Dev, Test & UAT Environments
- Impact on Storage Credits & Cost
- Clone + Time Travel Powerful Combination

DAY
13

Tasks & Change Data Capture

- Tasks Creation & CRON Scheduling Syntax
- Task Dependencies — Building DAG inside Snowflake
- Change Data Capture (CDC) Concepts
- INSERT / UPDATE / DELETE Handling in CDC
- Monitoring Task Execution History

DAY
14

Streams & Advanced Loading

- Streams — Standard, Append-Only & Insert-Only
- CDC Pipeline using Streams + Tasks Together
- Continuous Data Loading Patterns
- Semi-Structured Data — VARIANT, ARRAY, OBJECT Types
- FLATTEN & PARSE_JSON Functions for JSON Data

DAY
15

Data Masking & Security

- Dynamic Data Masking Policies — Setup & Apply
- Masking PII Columns — Real World Scenarios
- Row-Level Security — Row Access Policies
- Column-Level Security Best Practices
- Real-World Security Scenarios & Interview Q&A

DAY
16

Performance Tuning

- Clustering Keys — Manual & Automatic Clustering
- Micro-partition Pruning for Query Speed
- Query Profiling with EXPLAIN PLAN
- Result Cache & Metadata Cache Usage
- Query Optimization Techniques & Best Practices

DAY
17

UDFs & Stored Procedures

- SQL User Defined Functions (UDFs) — Creation & Use
- JavaScript UDFs for Complex Logic
- Stored Procedures in JavaScript & Snowflake Scripting
- Python Integration with Snowpark
- When to Use UDFs vs Stored Procedures

DAY
18

Views

- Standard Views — Creation & Use Cases
- Secure Views — Data Sharing Security

- Materialized Views — Setup & Refresh Strategies
- View Dependency Management & Tracking
- Performance Considerations for Each View Type

DAY
19

Slowly Changing Dimensions (SCD)

- SCD Type 1 — Overwrite (Simple Update)
- SCD Type 2 — History with Effective & Expiry Dates
- SCD Type 3 — Previous Value Column Approach
- MERGE Statement for SCD Implementation
- Practical Lab — Build SCD Pipeline End to End

DAY
20

Data Sharing

- Data Provider & Consumer Model — Architecture
- Creating Secure Shares in Snowflake
- Reader Accounts for Non-Snowflake Consumers
- Snowflake Marketplace — Data Exchange Overview
- Resume Preparation & LinkedIn / Naukri Profile Creation

DAY
21

Pricing & Cost Control

- Snowflake Pricing Model — Compute + Storage Explained
- Credits — What They Are & How They Are Consumed
- Cost Optimization Strategies for Real Projects
- Resource Monitors & Budget Alerts Setup
- Edition Differences — Standard, Enterprise, Business Critical

DAY
22

Date & Time Functions

- Date & Timestamp Data Types in Snowflake
- DATE_TRUNC, DATEADD, DATEDIFF Functions
- Time Zone Concepts & CONVERT_TIMEZONE Function
- Window Functions with Date Ranges
- Interview Questions on Date & Time Functions

DAY

Dynamic Tables & Best Practices

23

- Dynamic Tables — What They Are & Use Cases
- Target Lag Configuration & Refresh Strategy
- Dynamic Tables vs Materialized Views — Comparison
- Development Best Practices for Snowflake Projects
- 1st Mock Interview — Based on Completed Sessions

DAY
24

Real-Time Project Introduction

- Project Architecture Design — Medallion Pattern
- Bronze / Silver / Gold Layer Concept
- Business Requirement Analysis & Source Mapping
- Tools Setup — GitHub, Snowflake, DBT Cloud
- Project Kickoff & Environment Configuration

DAY
25

Project Lab

- Data Loading from S3 into Bronze Layer
- Streams & CDC Pipeline Setup
- Transformation Logic — Silver Layer Development
- Gold Layer — Reporting Tables Creation
- Real Interview Preparation — Start Attending Interviews

DAY
26

Final Lab & Snowflake Revision

- Complete Snowflake Architecture Revision
- All Key Concepts Summary & Mind Map
- Top 50 Snowflake Interview Questions & Answers
- Scenario-Based Interview Questions
- Q&A Session & Doubt Clearance

⚙️ PART B: DBT

Data Build Tool — Modern ELT Transformations
Days 27 – 36 | 10 Sessions

DAY
27

DBT Introduction

- What is ELT vs ETL — Modern Data Stack Shift
- DBT Cloud Setup & Account Creation
- Snowflake + DBT Integration & Connection Setup
- DBT Project Structure — Folders & Files Explained
- First DBT Run — Models, Compile & Execute

DAY
28

Models & Materializations

- DBT Models — Writing .sql Model Files
- Table Materialization — Creates Physical Tables
- Incremental Materialization — Process Only New Data
- Ephemeral Materialization — CTE Based (No Physical Object)
- View Materialization — Creates Views in Snowflake

DAY
29

Sources & Seeds

- Source Configuration in schema.yml File
- Referencing Sources in DBT Models with source()
- Seeds — Load CSV Reference Data into Snowflake
- Source Freshness Checks — Detect Stale Data
- Source Documentation & Testing

DAY
30

Tests & Snapshots

- Generic Tests — not_null, unique, accepted_values, relationships
- Singular (Custom) Tests — Write Your Own Test Logic
- Snapshots for SCD Type 2 Implementation
- Snapshot Strategy Configuration — Timestamp & Check
- Running Tests in CI/CD Pipeline

DAY
31

Macros & Jinja Templating

- Jinja Templating Basics in DBT — Variables & Loops
- Creating Reusable Macros for Common Logic
- Using ref() and source() Functions Effectively
- Dynamic SQL Generation with Macros
- Custom Schema & Database Override Macros

DAY
32

Hooks & Documentation

- Pre-Hook & Post-Hook Configuration in DBT
- On-Run-Start & On-Run-End Hooks
- Auto Documentation Generation with dbt docs generate
- DBT Docs Serve — Browse Documentation Locally
- Lineage Graph — Visual Data Flow in DBT

DAY
33

Analysis & Packages

- Analysis Folder — Ad-Hoc Queries in DBT Project
- DBT Packages — dbt_utils, dbt_expectations, codegen
- packages.yml Configuration & Installation
- Using Package Macros in Your Project
- Popular Community Packages & Best Practices

DAY
34

Deployment & Git Integration

- Git Version Control for DBT Projects
- Branching Strategy — Feature, Dev, Main Branches
- DBT Cloud Deployment Environments — Dev & Prod
- Scheduled Jobs in DBT Cloud
- CI/CD Integration — Auto Run on Pull Request

DAY
35

DBT Project Lab

- Real-Time DBT Transformations on Snowflake Data
- Staging Models — Clean Raw Data
- Intermediate Models — Business Logic

- Mart Models — Final Reporting Tables
- 2nd Mock Interview — Advanced + Scenario Questions

DAY
36

DBT Revision & Interview Prep

- Complete DBT Concept Revision
- Top 30 DBT Interview Questions & Answers
- Scenario-Based DBT Questions from Real Interviews
- DBT + Snowflake Combined Architecture Discussion
- Q&A Session & Doubt Clearance

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✈️ PART C: APACHE AIRFLOW

Pipeline Orchestration & Scheduling
Days 37 – 47 | 11 Sessions

Why Apache Airflow in This Course?

Airflow is the world's most popular pipeline orchestration tool — used by Airbnb, Uber, Netflix & 10,000+ companies globally. It is 100% FREE, open source, and works perfectly with both Snowflake and DBT. Adding Airflow makes this the most complete and most demanded Modern Data Engineering stack of 2026.

DAY
37

Python Basics for Airflow (No Prior Python Needed!)

- Why Python? Airflow uses Python only for simple configuration, NOT complex coding
- Variables & Data Types — strings, numbers, dates (15 mins of practice)
- Functions — def keyword, calling functions, passing values (20 mins)
- Import Statements — how to bring in libraries like datetime (10 mins)
- Indentation Rules — the only Python rule that really matters in Airflow
- Hands-on: Write your first 10-line Python script — confidence guaranteed!

DAY
38

Apache Airflow Introduction

- What is Pipeline Orchestration & Why It Matters
- Apache Airflow — The Scheduler + Manager of Data Pipelines
- Airflow Architecture — Scheduler, Webserver, Executor, Metadata DB
- Why Airflow with Snowflake + DBT — The Golden Trio
- Installation & Setup — Local (pip) & Docker Options

DAY
39

Core Concepts — DAGs & Tasks

- What is a DAG (Directed Acyclic Graph) — Workflow Definition
- Tasks — The Individual Steps Inside a DAG
- Operators — PythonOperator, BashOperator, SQLOperator
- Task Dependencies — set_upstream() & set_downstream()
- Airflow Web UI Walkthrough — Dashboard, DAG View, Logs

DAY
40

Operators Deep Dive

- PythonOperator — Run Any Python Function
- BashOperator — Execute Shell Commands
- EmailOperator — Send Notifications
- BranchPythonOperator — If/Else Logic in Pipelines
- Hands-on Lab: Build Multi-Step Pipeline with Different Operators

DAY
41

Fivetran Introduction — BONUS: Industry Tool Awareness

- What is Fivetran? — Automated Data Ingestion with ZERO coding required
- Fivetran Architecture — Source → Connector → Destination (Snowflake)
- 300+ Ready-Made Connectors — Salesforce, HubSpot, Google Ads, MySQL & more
- Fivetran Free Trial Account Setup & Dashboard Walkthrough
- Hands-on: Connect Google Sheets → Snowflake using Fivetran (free connector!)
- Sync Types — Full Refresh vs Incremental Sync with real world examples

DAY
42

Fivetran vs Airflow — When to Use Which?

- Side-by-Side Comparison — Fivetran (no-code) vs Airflow (code-based) ingestion
- When to choose Fivetran — SaaS sources like Salesforce, HubSpot, Google Analytics
- When to choose Airflow — custom logic, complex pipelines, full pipeline control
- Can they work TOGETHER? YES! Fivetran ingests → Airflow orchestrates DBT runs
- Real Company Scenarios — Startup uses Fivetran, MNC uses Airflow — why?
- Top Interview Questions on both Fivetran & Airflow ingestion approaches

DAY
43

Snowflake + Airflow Integration

- SnowflakeOperator — Connect Airflow to Snowflake
- Setup Snowflake Connection in Airflow Connections UI
- Run Snowflake SQL Queries from Airflow DAG
- Load Data into Snowflake via Airflow Pipeline
- Hands-on Lab: Full Snowflake Pipeline Orchestrated by Airflow

DAY
44

DBT + Airflow Integration

- DBT Operator Setup in Airflow
- Trigger DBT Models from Airflow DAG
- Schedule DBT Runs Automatically with CRON
- Handle DBT Failures — Retry & Alert Strategy
- Hands-on Lab: Airflow Triggers DBT → DBT Transforms Snowflake

DAY
45

Advanced Airflow Features

- XComs — Pass Data Between Tasks
- Airflow Variables & Connections — Centralized Config
- CRON Scheduling Patterns — Daily, Weekly, Hourly
- Task Retries, Timeouts & SLA Monitoring
- Branching & Conditional Logic in Complex Pipelines

DAY
46

Monitoring, Alerts & Best Practices

- Airflow Dashboard Deep Dive — Monitoring Pipeline Health
- Log Analysis — Debugging Failed Tasks
- Email Alerts on Pipeline Failure Setup
- Airflow Best Practices for Production Pipelines
- Common Airflow Interview Questions & Answers

DAY
47

Capstone Project — Full Pipeline

- End-to-End Pipeline: Source → Airflow → Snowflake → DBT
- Airflow Orchestrates the Entire Data Journey
- Error Handling & Retry Logic in Real Project
- Project Documentation & Architecture Diagram
- Final Mock Interview — All 3 Technologies Combined

WHY RISHAB INFORMATICA GROUP?

Your Trusted Partner for Data Engineering Career Success

☐ Limited Batch Size

Max 25 students — personal attention guaranteed

☐ No Mute Policy

100% interactive — ask questions anytime

☐ LMS Access

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☐ 2 Real Projects

End-to-end real-time project experience

☐ Interview Focused

Interview prep from Day 1 of training

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